



Dominique Makowski

ASSISTANT PROFESSOR IN PSYCHOLOGY

University of Sussex, Brighton, UK

[✉ D.Makowski@sussex.ac.uk](mailto:D.Makowski@sussex.ac.uk) | [🌐 realitybendinglab.com](http://realitybendinglab.com)

[🎓 Google Scholar](#)

LANGUAGES: FRENCH · POLISH · ENGLISH · ITALIAN (B1)

*Trained as a clinical neuropsychologist and CBT psychotherapist, I am an assistant professor at the University of Sussex, where I head the [Reality Bending Lab](#). My research has three strands. **Reality and its distortions:** visual illusions, deception and lie detection, misinformation and fake news, beliefs about AI-generated content, fiction, and altered states of consciousness. **The body in cognition:** how interoception, physiological states and emotion regulation shape what we take to be real. **Methods:** Bayesian statistics, computational models and psychophysiology, as well as the development of new software, measures and paradigms, released as open source (easystats, NeuroKit, cogmod) to support open and rigorous science.*

Experience



Assistant Professor (Current Position)

SCHOOL OF PSYCHOLOGY, UNIVERSITY OF SUSSEX

[UK](#)

2023 - Present

- Full teaching service on courses including Bayesian Statistics and Psychophysiology.
- Head of the [Reality Bending Lab](#): two doctoral students to date, around two full-time placement students and a dozen dissertation students a year, funded summer interns and research assistants.
- School-level leadership responsibilities, including research ethics, the research participation scheme and postgraduate programme development.



Research Fellow

SCHOOL OF SOCIAL SCIENCES, NANYANG TECHNOLOGICAL UNIVERSITY (NTU)

[Singapore](#)

2019 - 22

- Awarded the NTU Presidential Postdoctoral Fellowship (2021 - 22), a competitive two-year grant bridging postdoc and faculty. Led an independent research track on the neurocognitive and affective processes underlying beliefs about reality within the [Clinical Brain Lab](#), managing a team of three research assistants and a research budget of roughly USD 200,000.
- Research Fellow (2019 - 20) on a Singapore Ministry of Defence funded project on the neurocognitive correlates of deception: study design, data collection, analysis and publications; co-supervision of one PhD student and two research assistants.

Education



Doctor of Philosophy (PhD) - Psychology

UNIVERSITÉ PARIS CITÉ

[France](#)

2014 - 18



Certified Psychotherapist - Cognitive Behavioural Therapy (CBT)

FRENCH ASSOCIATION FOR COGNITIVE BEHAVIOURAL THERAPY (AFTCC)

[France](#)

2014 - 17



BSc, MSc - Clinical Neuropsychology and Cognitive Psychology

UNIVERSITÉ PARIS CITÉ

[France](#)

2009 - 14

Clinical Practice

Assistant Neuropsychologist

DEPARTMENT OF ADULT PSYCHIATRY, PITIÉ-SALPÊTRIÈRE HOSPITAL (AP-HP)

[Paris, France](#)

2013 - 14

- Professional placement (580 h): neuropsychological assessment and diagnosis of inpatients with mood disorders, psychosis and attention deficit hyperactivity disorder, and cognitive remediation.

Assistant Neuropsychologist

DEPARTMENT OF GERIATRIC PSYCHIATRY, SAINTE-ANNE HOSPITAL

[Paris, France](#)

2012 - 13

- Professional placement (304 h): assessment and diagnosis of complex presentations combining psychiatric disorders with early dementia in older adults.

Assistant Neuropsychologist

DEPARTMENT OF GERONTOLOGY, LA PORTE VERTE HOSPITAL

[Versailles, France](#)

2012

- Observation placement (96 h): assessment of memory deficits in neurodegenerative disease, somatic ageing-related illness and depression.

Grants and Awards

GRANTS

Illusion Sensitivity and Phenomenological Control	£20,000
CENTRE FOR OPEN SCIENCE · REGISTERED REPORTS IN CONSCIOUSNESS RESEARCH · PI · 2024 - 26	
Perception of Real vs. Fake Information	£655,000
TEMASEK LABORATORIES, NTU · BEHAVIOURAL AND COMMUNICATION SCIENCE PROGRAMME · CO-PI · 2022 - 25	
Crowdedness and Environment Typology (NICE)	£179,000
SINGAPORE NATIONAL RESEARCH FOUNDATION · INTRA-CREATE GRANT · CO-PI · 2021 - 22	
“That’s Fake News”: Belief in Deceptive Information	£258,000
NANYANG TECHNOLOGICAL UNIVERSITY · PRESIDENTIAL POSTDOCTORAL FELLOWSHIP · PI · 2020 - 22	

AWARDS

SIPS Commendation	2024
SOCIETY FOR THE IMPROVEMENT OF PSYCHOLOGICAL SCIENCE For the NeuroKit Python package.	
Openness in Research Award	2024
UNIVERSITY OF SUSSEX RESEARCH CULTURE AWARDS For building software that makes open research accessible, and for running fully transparent studies.	
School of Psychology Award	2024
UNIVERSITY OF SUSSEX For outstanding student supervision.	
Mission Award	2023
SOCIETY FOR THE IMPROVEMENT OF PSYCHOLOGICAL SCIENCE For the easystats collection of R packages.	
Best Article Award	2020
PSYCHONOMIC SOCIETY For the fictional reappraisal paper, the main output of the PhD.	
SIPS Commendation	2020
SOCIETY FOR THE IMPROVEMENT OF PSYCHOLOGICAL SCIENCE For the bayestestR R package.	

Teaching

Areas covered, grouped rather than listed module by module: lectures, seminars and practicals since 2013 at the University of Sussex and Université Paris Cité, alongside invited doctoral workshops.

METHODS AND STATISTICS

Bayesian Statistics	MSc · PhD	The Bayesian framework, estimation and inference; multilevel and cognitive models with brms. Module convenor ; designed from scratch. UNIVERSITY OF SUSSEX · 2024 - PRESENT
Advanced Statistics	BSc · MSc · PhD	Regression and mixed models; psychometrics; experimental methodology and scientific reasoning. Taught in R throughout. UNIVERSITY OF SUSSEX · AN ADVENTURE IN STATISTICS · 2023 - PRESENT · QUALITATIVE AND QUANTITATIVE METHODS · 2023 - 25 · UNIVERSITÉ PARIS CITÉ · 19 H · 2017 - 18
Programming and Tools	MRes · PhD	Python for experiment building and signal processing; R; version control and collaboration with Git and GitHub; agentic coding tools for research. UNIVERSITY OF SUSSEX · MODERN RESEARCH WORKFLOWS: GITHUB · 2025 · UNIVERSITY OF ZURICH · 24 H · 2022

COGNITION AND PSYCHOBIOLOGY

Psychophysiology	BSc · MSc	Signal acquisition and preprocessing; heart rate variability; electrodermal activity; interoception and interoceptive models; embodied cognition. UNIVERSITY OF SUSSEX · COGNITIVE NEUROSCIENCE · TOPICS IN NEUROSCIENCE · 2023 - PRESENT · UNIVERSITY OF BASEL · 8 H · 2024
Cognitive Neuroscience	BSc	Neuroanatomy; memory; emotion regulation; neuroimaging; mindfulness and meditation. UNIVERSITÉ PARIS CITÉ · 90 H · 2017 - 18
Cognitive Psychology	BSc	Models of working memory; mental imagery and categorisation. UNIVERSITÉ PARIS CITÉ · 48 H · 2014 - 16

INDIVIDUAL DIFFERENCES AND CLINICAL PSYCHOLOGY

Individual Differences	BSc	Intelligence; personality; creativity. UNIVERSITÉ PARIS CITÉ · 144 H · 2015 - 17
Clinical Neuropsychology	BSc	Neuropsychological syndromes; neuropsychological examination, diagnosis and psychometrics; neuropsychology in psychiatry. UNIVERSITÉ PARIS CITÉ · 51 H · 2017 - 18
Rehabilitation and Practice	MSc	Cognitive rehabilitation and intervention; neuropsychology and disability; neurodegenerative and neurodevelopmental disorders; clinical practice supervision. UNIVERSITÉ PARIS CITÉ · 29 H · 2017 - 18

CONVENING AND CURRICULUM

Teaching Qualifications	FHEA
ADVANCE HE · 2025 · UNIVERSITÉ PARIS CITÉ · 2017	
Fellow of the Higher Education Academy (FHEA, 2025; Associate Fellow 2024), assessed against the UK Professional Standards Framework. Academic Teaching Certificate in pedagogy and innovative teaching (40 h, 2017).	
Bayesian Statistics	2024 -
UNIVERSITY OF SUSSEX · MODULE CONVENOR New master’s and PhD module, designed from scratch .	
Invited instruction	2021 - 25
BASEL · ZURICH · PARIS · SINGAPORE · OHBM Doctoral workshops on psychophysiology, Python for signal processing, Bayesian statistics and mixed models, and an OHBM educational course .	

SUPERVISION AND EXAMINING

Doctoral supervision and examining	2019 -
UNIVERSITY OF SUSSEX · NTU SINGAPORE PhD supervision at Sussex and co-supervision at NTU. Internal examiner for four Sussex PhD theses , on consciousness science, interoception and quantitative methods.	
Dissertation and placement students	2023 -
UNIVERSITY OF SUSSEX Around nine undergraduate and three master’s dissertation students a year , two full-time placement students, and funded summer interns.	
Research staff	2019 -
REALITY BENDING LAB · CLINICAL BRAIN LAB More than seven full-time research assistants , several of whom went on to funded PhD positions.	

Software and Tools

SOFTWARE



easystats **41.2M downloads**
R · CO-FOUNDER AND CORE MAINTAINER · 2019 - PRESENT
Ten R packages for the post-processing of statistical models: parameters, effect sizes, diagnostics, Bayesian indices, marginal effects, reporting. **Used in over 1,000 publications. SIPS Mission Award (2023).**



NeuroKit **130k downloads/month**
PYTHON · CREATOR AND LEAD MAINTAINER · 2017 - PRESENT
Signal processing for ECG, PPG, EDA, EMG, EOG and EEG, from peak detection to heart rate variability and complexity indices. **2,355 stars** and dozens of contributors. **SIPS Commendation (2024).**

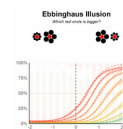


Pyllusion **82 stars**
PYTHON · CREATOR AND MAINTAINER · 2020 - PRESENT
Generates visual illusions parametrically, separating illusion strength from the objective difference between the targets, so that sensitivity can be measured continuously.



cogmod
R · SOLE AUTHOR · 2024 - PRESENT
Bayesian models of reaction times, choices and ratings - drift diffusion, linear ballistic accumulator, racing diffusion - as custom brms families.

MEASURES AND PARADIGMS



Illusion Game
SCIENTIFIC REPORTS · 2023
Parametric task measuring visual illusion sensitivity as a continuous trait, with evidence for a general factor.



LIE Scale
CURRENT PSYCHOLOGY · 2021
Sixteen-item measure of lying styles and attitudes: ability, frequency, negativity and contextuality.



MINT Scale
PSYARXIV PREPRINT
Multimodal interoception questionnaire, across cardiac, respiratory, gastric and other bodily channels.

DoggoNogo
IN DEVELOPMENT

Service and Leadership

SCHOOL AND UNIVERSITY

MSc Advanced Computational Psychology **2026 -**
UNIVERSITY OF SUSSEX · PROGRAMME DEVELOPER
Design of a **new postgraduate programme** from scratch: curriculum structure, module content and learning outcomes.

Academic Lead, Research Participation Scheme **2025 -**
UNIVERSITY OF SUSSEX · SCHOOL OF PSYCHOLOGY
Redesigned the credit allocation and study approval rules to be fairer and more transparent, and trialled changes raising both the educational value of participation and the quality of the data.

Cross-Schools Research Ethics Committee **2023 -**
UNIVERSITY OF SUSSEX · MEMBER

Organiser, MORSE Seminar Series **2024 -**
CENTRE FOR OPEN SCIENCE AND RESEARCH REFORM, SUSSEX
Methods for Open and Reproducible Science: a seminar series on openness, reproducibility and replicability in psychological science.

EDITORIAL AND PEER REVIEW

Statistical Editor **2025 -**
ROYAL SOCIETY OPEN SCIENCE

Associate Editor **2020 - 24**
FRONTIERS IN PSYCHOLOGY · QUANTITATIVE PSYCHOLOGY AND MEASUREMENT

Regular Reviewer **2018 -**
JOURNAL OF OPEN SOURCE SOFTWARE

Reviewer **2017 -**
NATURE HUMAN BEHAVIOUR · BIOLOGICAL PSYCHIATRY · SCIENTIFIC REPORTS
Also Acta Psychologica, Journal of Experimental Psychology: Human Perception and Performance, Neuropsychologia, Personality and Individual Differences and others.

Engagement and Knowledge Exchange

PUBLIC ENGAGEMENT AND OUTREACH

Artificial Emotion Symposium **2026**
ESCAN · SAPIENZA UNIVERSITY, ROME · CO-ORGANISER
Four-talk symposium on affective, physiological and neural responses to artificial intelligence, for the European Society for Cognitive and Affective Neuroscience.

Open Science Champion **2024**
BRITISH NEUROSCIENCE ASSOCIATION · MEMBERS' MEETING
Invited to argue the open science position in the meeting's interactive debate, alongside a Sustainability Champion and Wellcome.

Lab Blog **40+ posts**
REALITY BENDING LAB · 2019 -
Methods tutorials, research explainers, advice for students and essays on research culture, written for a general and early-career audience.

Media and Public Talks **2015 -**
TELEVISION · PODCAST · PINT OF SCIENCE
French national television on violent images, body and brain; the *Learning Bayesian Statistics* podcast; Pint of Science; a public debate on sensory illusions in Bordeaux; press coverage of the 3D-cinema memory and AI-generated faces studies.

KNOWLEDGE EXCHANGE AND CONSULTANCY

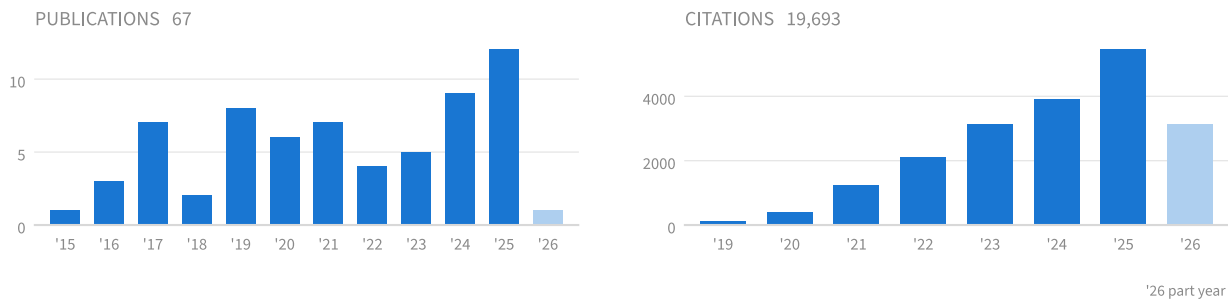
Scientific Advisor **2025 -**
ARTIFICIAL INTELLIGENCE START-UP, SINGAPORE · CLIENT CONFIDENTIAL
Scientific advising on an applied project on deception detection.

Contract Research **2026**
FRAGRANCE INDUSTRY · CLIENT CONFIDENTIAL
Contract research on the relationship between smells and emotions.

Contracted Instructor **2026**
SOUTH EAST DOCTORAL TRAINING ARC (SEDARC)
Reusable *Introduction to GitHub* lecture for PhD students across the six universities of the partnership.

Contracted Developer **2025**
INSTITUTE FOR ADVANCED CONSCIOUSNESS STUDIES
Hyperscanning capabilities for open-source portable electroencephalography (EEG) software.

Research Impact



- Most cited researcher in the School of Psychology, University of Sussex, since 2024.
- *h*-index 27, i10-index 38; 40 of 67 publications as first (29) or senior (11) author.
- Three contributions have their own Wikipedia article: the [probability of direction](#) (pd), [easystats](#) and [NeuroKit](#).

GOOGLE SCHOLAR, JUNE 2026